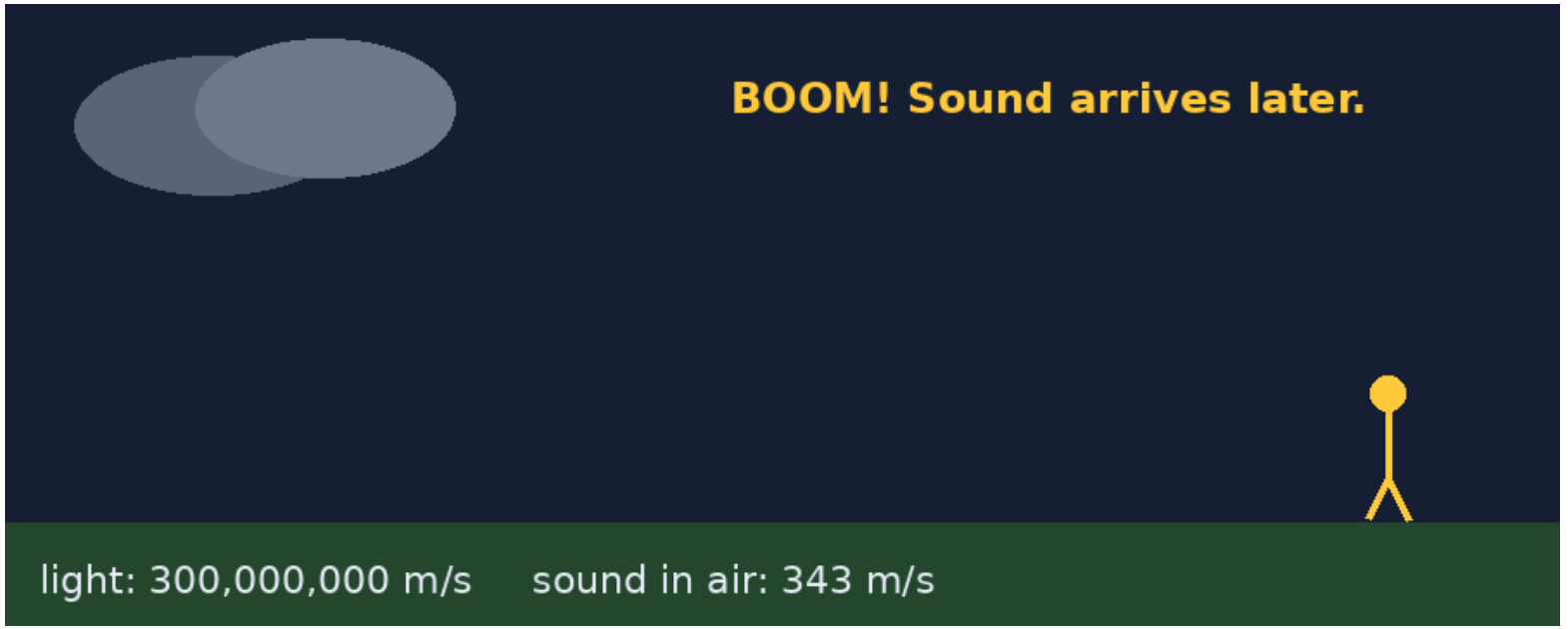
4.05 Waves and Their Medium

How does the medium change the speed of a wave?



Today's Goals

I can explain why we see lightning before we hear thunder.

I can find speed with speed = distance ÷ time.

I can explain why sound travels faster in warm air and water.

I can compare sound speed in solids, liquids, and gases.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- Mechanical waves need a medium. The medium moves, then returns to its rest position.
- The speed of a wave depends on the material (sound travels faster through most solids than gases).
- Changing the medium, like its temperature, changes the wave speed.

Inquiry skills (Grades 5–8): "Use appropriate mathematics, tools and techniques to gather data" and "Think critically and logically to connect evidence and explanations" (today's Wave Speed and Temperature Lab).

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

In this lesson

- Integer warm-up
- Vocab check-in
- Wave Speed and Temperature Lab
- Counting thunder
- Hot air vs. cold air
- Solids, liquids, and gases
- Graph reveal: sound in water

CASE #2 • TUESDAY

Did the rubber duck drop 5 meters?

WAVE WORD: *Medium*

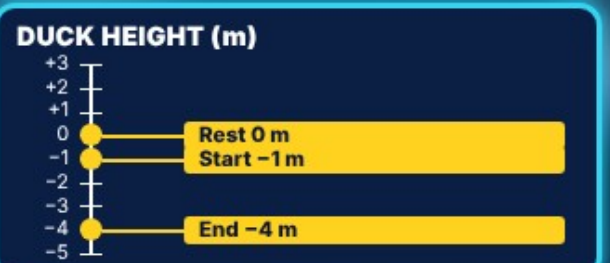
The material a wave travels through.

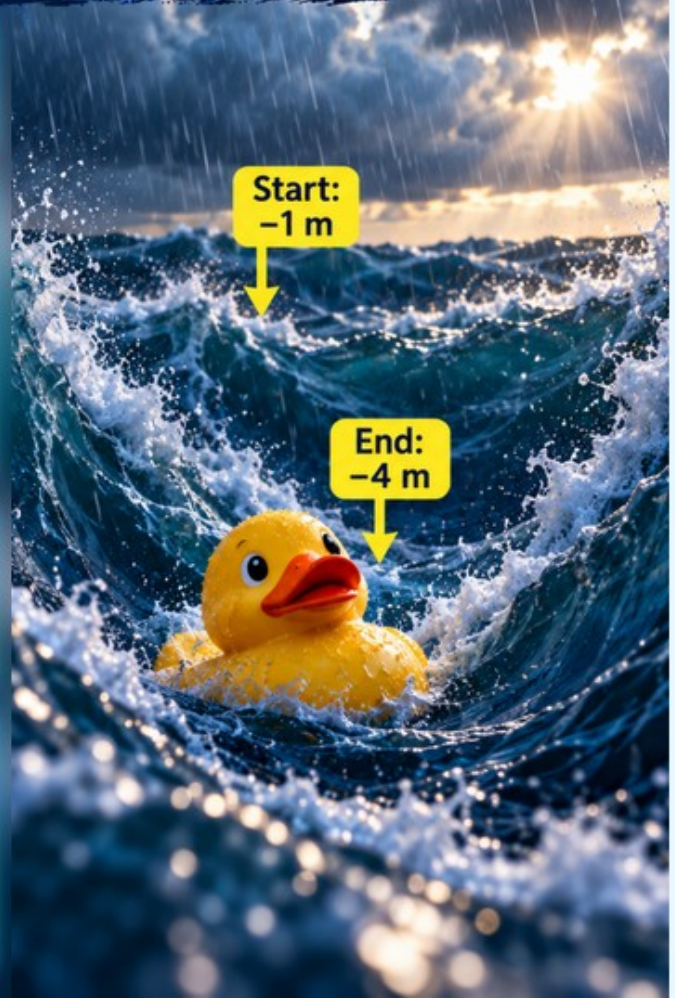
MYSTERY

A rubber duck floats on stormy Lake Erie. The water is the medium, so the duck bobs up and down as waves pass instead of traveling with them. The duck sits 1 meter below rest position. Then a deep trough drops it to 4 meters below rest position. Did the duck drop 5 meters?

CLUES

- Start: −1 m
- End: −4 m
- Change = end − start





CLAIM

Did the duck drop 5 meters?

☐ YES ☐ NO

EVIDENCE

Solve:

−4 − (−1) = _____

Rewrite as adding: −4 + _____ = _____

REASONING

Finish the sentence:

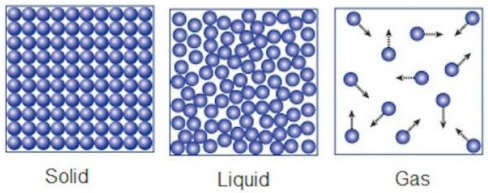
−4 − (−1) is the same as

−4 + _____ = _____.

So the duck dropped _____ m.

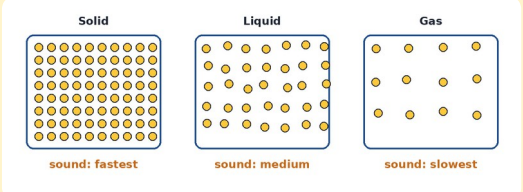
Vocab Check-In

From your Unit 4 vocabulary pretest and the 4.05 page. Starred words are on the pretest.



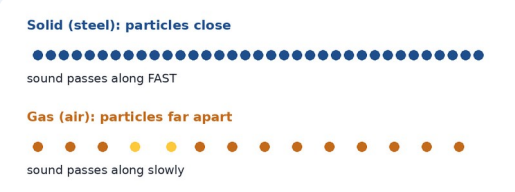
medium ★

the material a wave travels through (solid, liquid, or gas)



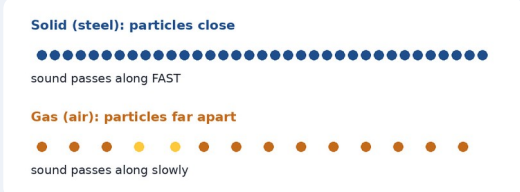
density ★

the amount of mass in a given volume



wave speed

how fast a wave travels through a medium



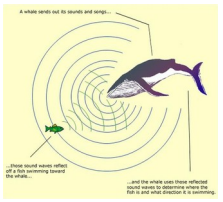
elasticity

how well a material springs back to its shape; stiff steel passes waves fast



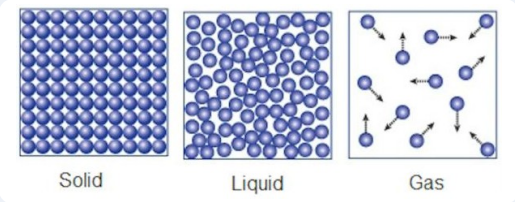
refraction ★

the bending of waves when they enter a new medium



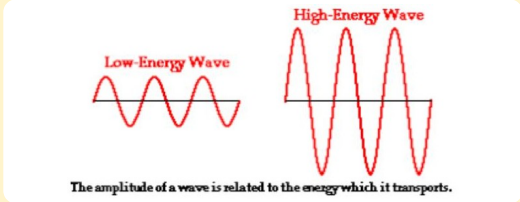
reflection

when waves bounce back after hitting a surface



absorption

when wave energy is taken in by the medium



energy ★

the ability to do work or cause change

Classkick: 4.05 Part 1 · Link 1

Watch First: Speed of Sound and Temperature



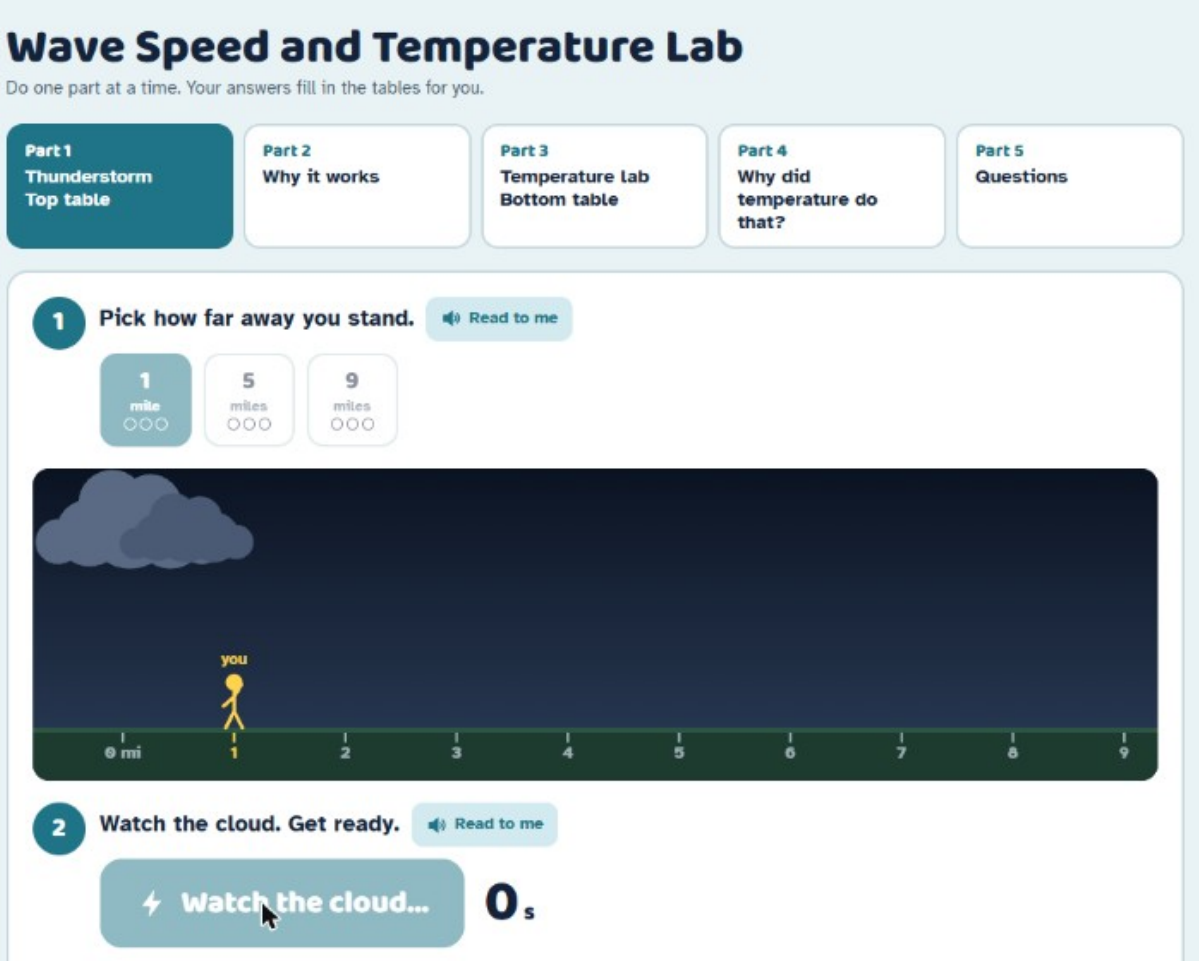
Plays here in Slide Show (needs internet)

Listen for:

- How fast does sound travel in air?
- Does sound go faster in warm or cold air?
- Why does temperature matter?

▶️ Backup: YouTube

Lab Part 1: Count the Thunder



1. Pick 1, 5, or 9 miles.
2. Click Make lightning. Watch the cloud for the flash.
3. Click I hear thunder right when you hear the BOOM.
4. Do each distance 3 times. The Top table fills in for you.
5. Copy your Top table onto your Classkick.

▶️ **Open the Wave Speed and Temperature Lab**

Find the Speed of Sound

speed = distance ÷ time

Distance	Your average time	Speed
1 mile = 1,609 m	4.7 s	1,609 ÷ 4.7 ≈ ?
5 miles = 8,047 m	23.5 s	?
9 miles = 14,484 m	42 s	?

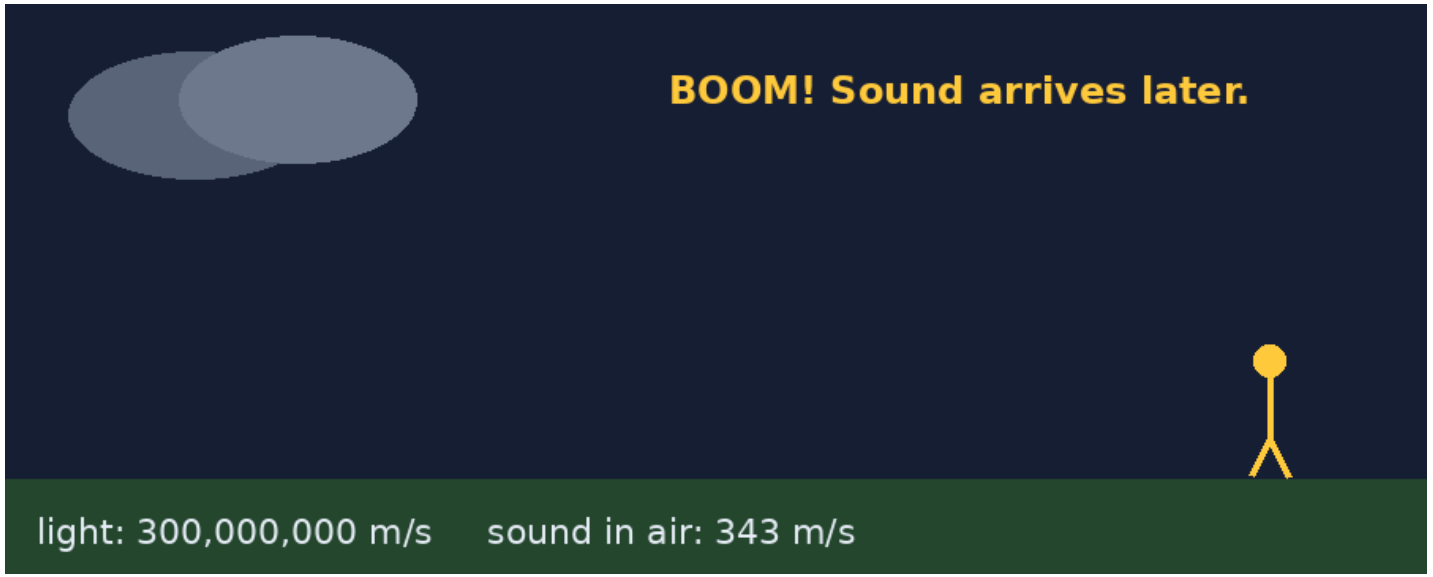
Try the 1-mile row together:

1,609 m ÷ 4.7 s ≈ **343 m/s**

Now try 5 miles and 9 miles. What do you notice about the speed?

Farther away = more TIME, but the SPEED stays the same.

Lab Part 2: Why Does It Work?



Light

an electromagnetic wave
about 300,000,000 m/s
does NOT need air

Sound

a mechanical wave
about 343 m/s in air
NEEDS a medium (air bumps air)

Lightning makes light and sound at the same time. Light gets to you almost instantly, so you see the flash first.



Use It in a Real Storm

Sound goes about 1 mile every 5 seconds. Count the seconds, then divide by 5.

You count 15 seconds

15 ÷ 5 = ___ miles away

You count 10 seconds

10 ÷ 5 = ___ miles away

You count 30 seconds

30 ÷ 5 = ___ miles away



Stay safe: When thunder roars, go indoors. Wait 30 minutes after the last thunder.

Lab Part 3: Does Temperature Matter?

Do one distance at a time. Click each temperature, then click Save. Here is the 1-mile row:

Distance	Time @ 20 °F	Speed @ 20 °F	Time @ 60 °F	Speed @ 60 °F	Time @ 100 °F	Speed @ 100 °F
1 mile	4.9 s	327.4 m/s	4.7 s	340.7 m/s	4.5 s	354.1 m/s
5 miles	?	?	?	?	?	?
9 miles	?	?	?	?	?	?

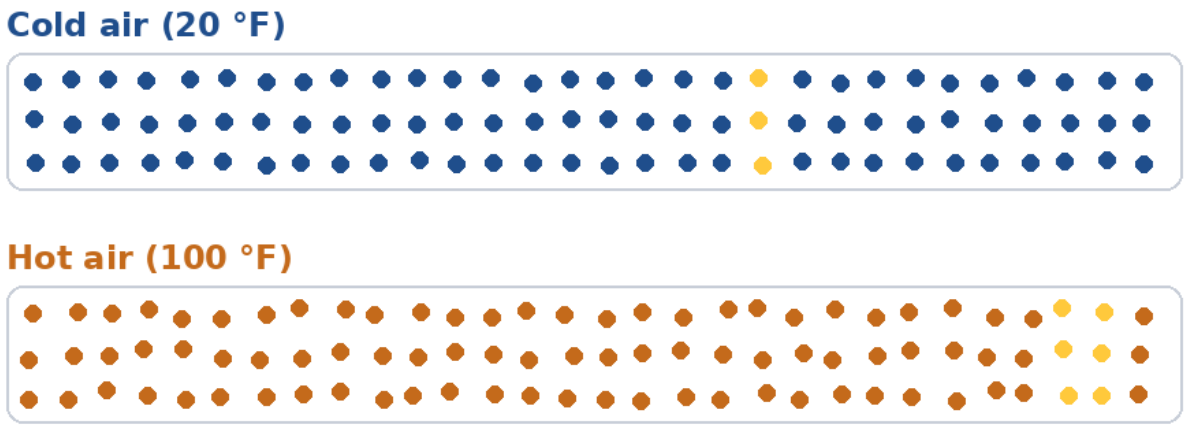
Think: Look across a row. Which temperature made the thunder arrive soonest? A shorter time means a faster speed.

▶ **Open the lab: Part 3**

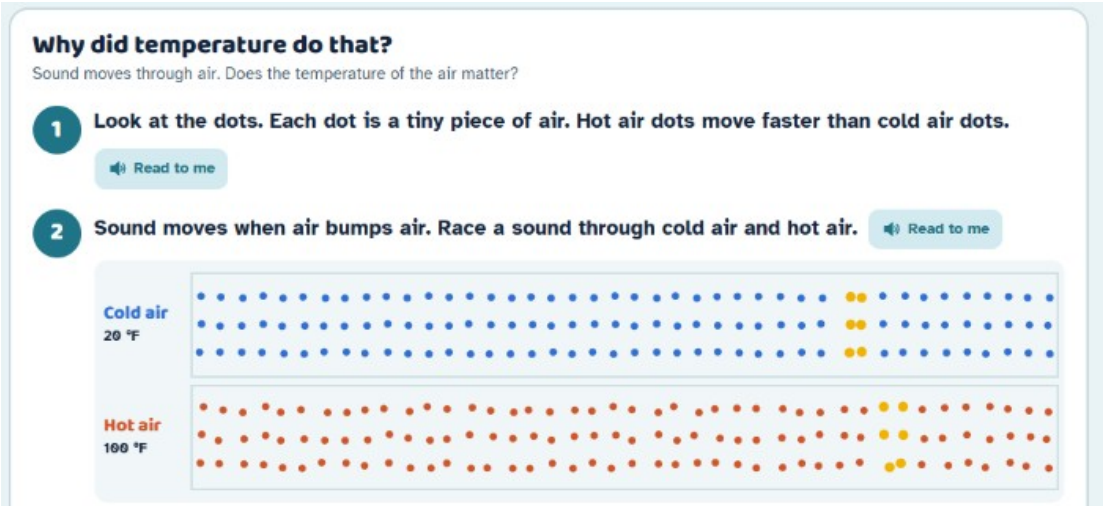
Not done? This part can be finished Friday.

Classkick: 4.05 Part 2 · Q5

Lab Part 4: Why Did Temperature Do That?



Animation: watch the yellow sound wave race.



Your lab: click Start the race.

Hot air particles move faster, so they bump into each other and pass the sound along faster.

Classkick: 4.05 Part 2 · Q6

Lab Part 5: Questions

1

Which is faster?

a light wave / a sound wave

4

Sound travels faster in...

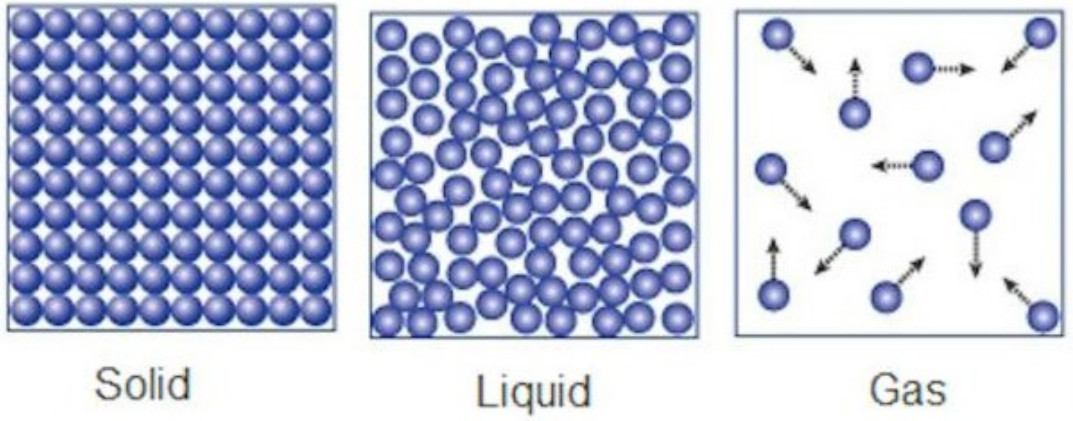
warm air / cold air

5

If you only change the medium (the air), the wave speed...

changes / stays the same

Solids, Liquids, and Gases



Solid (steel): particles close



sound passes along FAST

Gas (air): particles far apart



sound passes along slowly

Medium	Speed of sound
air (gas)	about 343 m/s
water (liquid)	about 1,480 m/s
steel (solid)	about 5,960 m/s

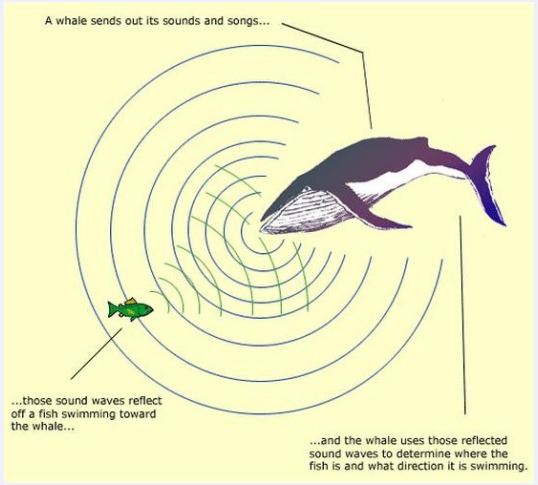
Ear on the track! Sound travels about 17 times faster through steel rails than through air, so you hear the train coming through the track first.

At a New Medium, a Wave Can...



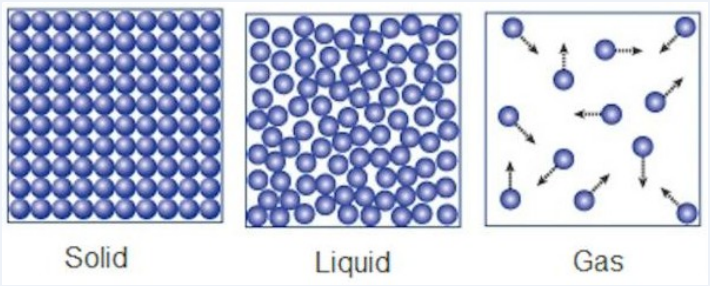
Refract (bend)

The pencil looks broken because light bends going from water to air.



Reflect (bounce back)

A whale's sound bounces off a fish. The echo tells the whale where the fish is.

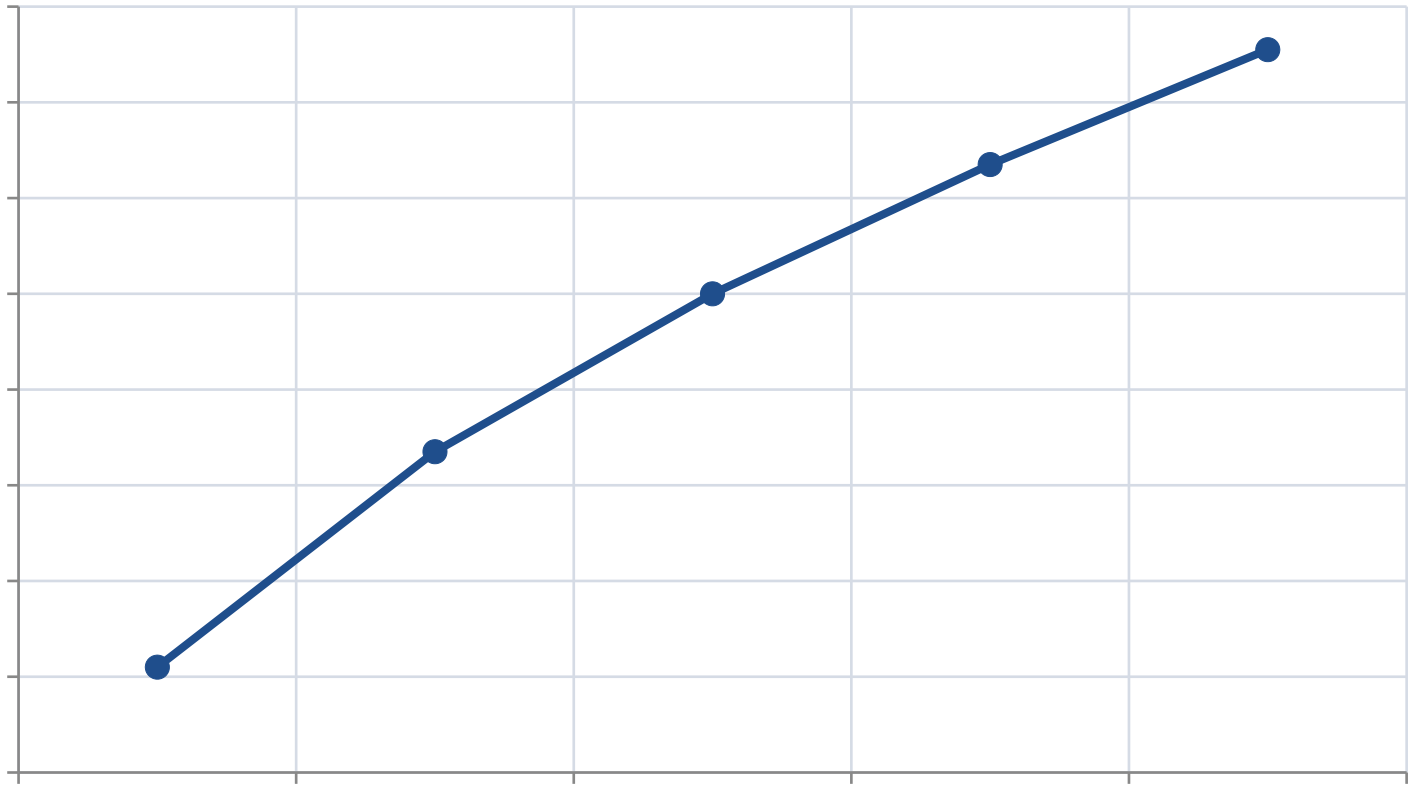


Be absorbed (taken in)

The medium takes in some of the energy, so sound gets fainter over distance.

Graph Reveal

Classkick: 4.05 Part 3 · Q7–Q10



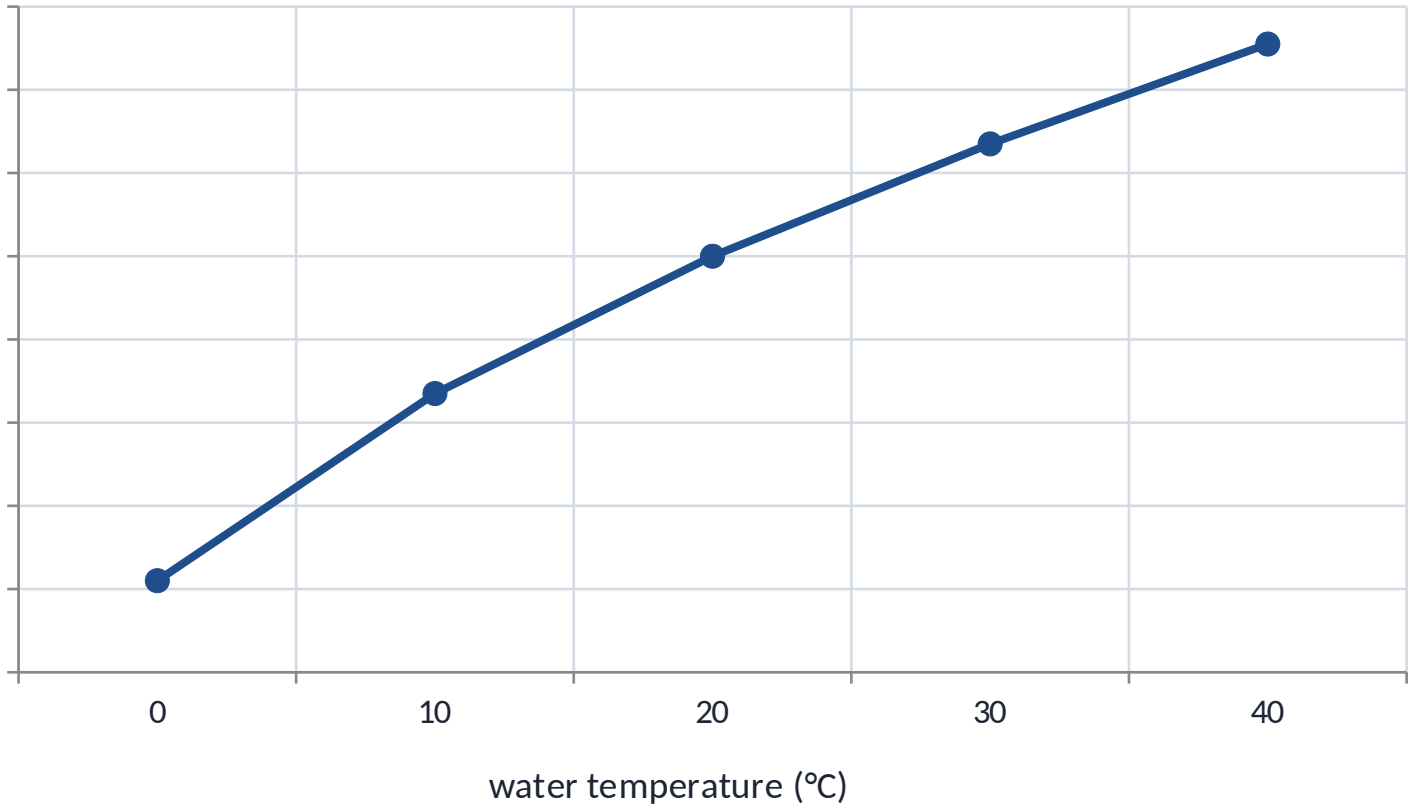
Step 1: What do you notice about the shape of the line?

No labels yet! Just look at the shape.

▶️ Open the graph reveal

Graph Reveal

Classkick: 4.05 Part 3 · Q7–Q10



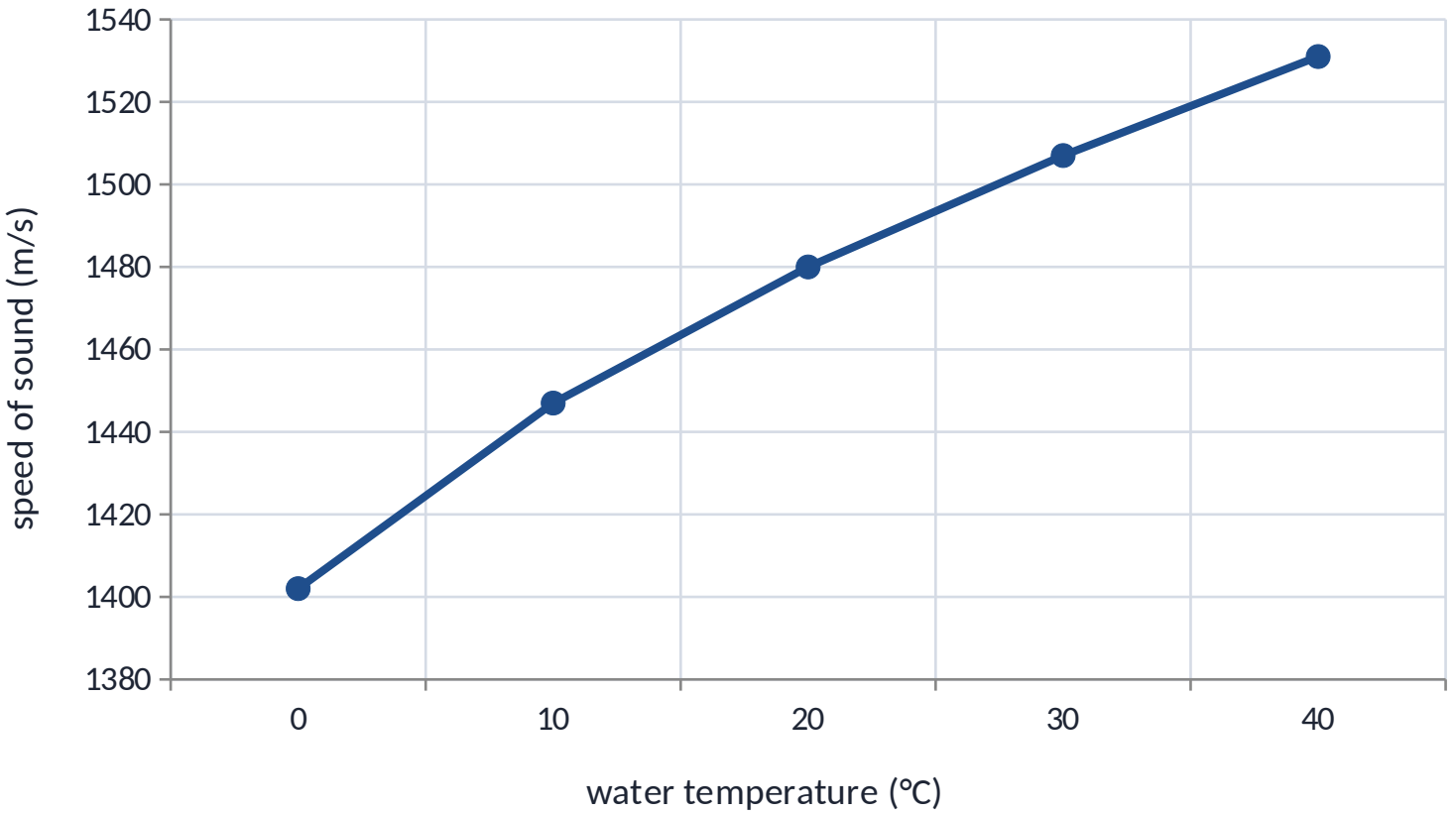
Step 2: What does the x-axis show?

Read the bottom of the graph.

▶️ Open the graph reveal

Classkick: 4.05 Part 3 · Q7–Q10

Graph Reveal



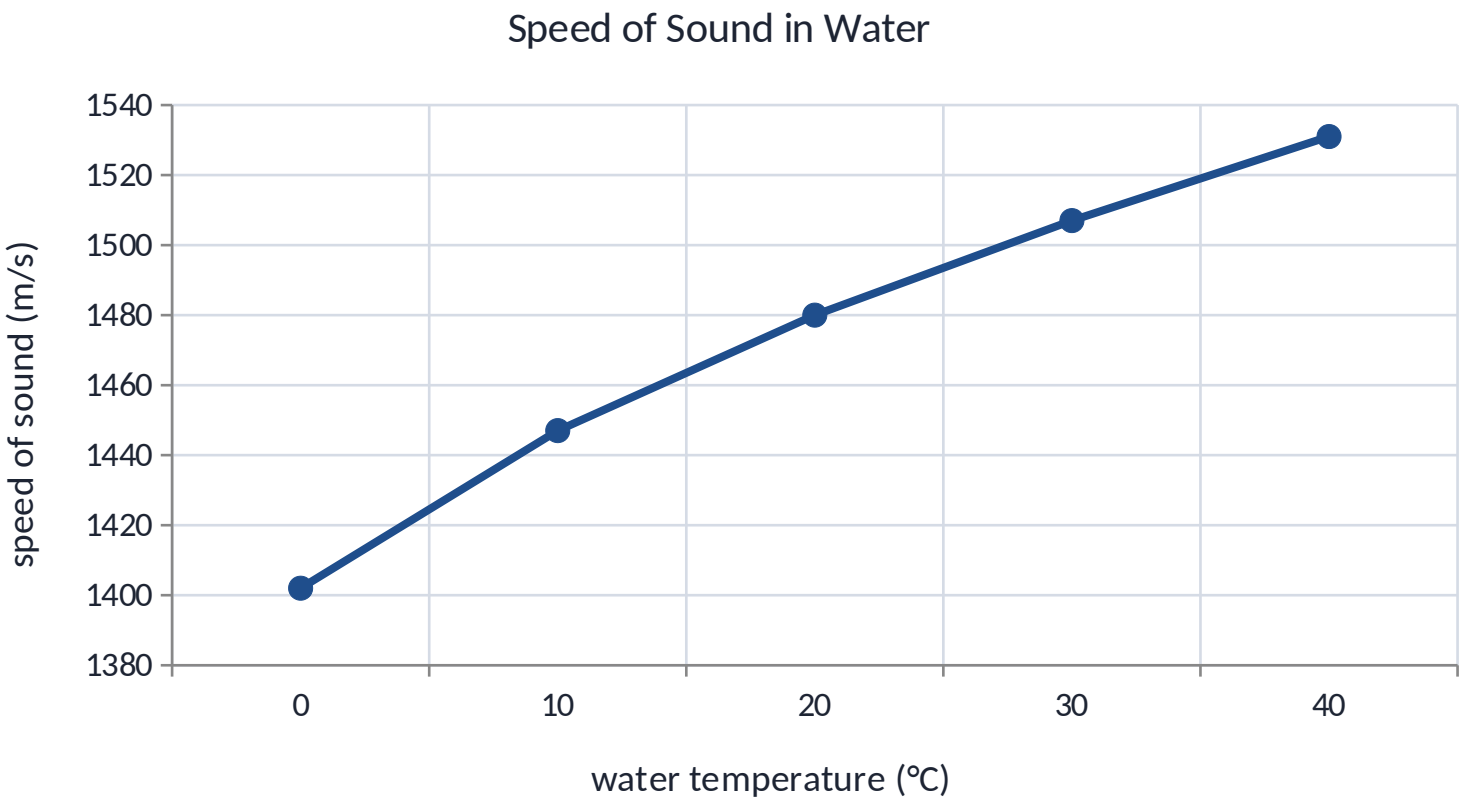
Step 3: What does the y-axis show?

Read the side of the graph.

▶️ Open the graph reveal

Graph Poll

Classkick: 4.05 Part 3 · Q7–Q10



Step 4: What is the pattern? Answer the poll!

Single choice poll

A. As water warms, sound gets faster.

Single choice poll

B. As water warms, sound gets slower.

Single choice poll

C. Temperature does not change sound speed.

Single choice poll

D. Sound is slower in water than in air.

Who Is Right?

Two hikers clap at the same canyon wall, one on a 35 °C summer day and one on a −5 °C winter day. Whose echo comes back sooner?

Ava says:

"The winter echo, because cold air is heavier and pushes sound faster."

Ben says:

"The same time, because air is air no matter the temperature."

Cam says:

"The summer echo, because warm air particles move faster and pass sound faster."

Dee says:

"The winter echo, because sound travels better when it is quiet and cold."

Single choice poll

A. Ava

Single choice poll

B. Ben

Single choice poll

C. Cam

Single choice poll

D. Dee

Key Ideas

1. Light arrives first: about 300,000,000 m/s. Sound in air is about 343 m/s.
2. speed = distance ÷ time. Distance changes the time, not the speed.
3. Sound is faster in warm air and warm water: the particles pass it along faster.
4. Sound is fastest in solids, slower in liquids, and slowest in gases.

Re-rate your goals: 1 = Not yet, 2 = Getting there, 3 = I can teach it